

Project: Expectation Decider.

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First 8 rows:

student id	study hours	attendance	group-disc	previous test	final exam pass
1	11.2	82	Yes	31	Pass
2	5.8	95	No	31	Fail
3	13.0	79	Yes	46	Pass
4	13.8	86	No	48	Pass
5	2.2	53	No	34	Fail
6	4.8	77	Yes	39	Fail
7	10.5	68	Yes	25	Pass
8	8.7	63	Yes	44	Fail

study hours attendance previous test score.

count	200.00	200.00	200.00
mean	9.88	78.06	44.16
std	3.53	11.88	12.07
min	1.50	47.00	12.00
25%	7.40	70.00	37.00
50%	9.80	79.00	43.00
75%	12.12	86.00	52.00
max	21.70	100.00	73.00

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Questions:

1 Understanding the Basics

What is Probability?

Probability measures how likely an event is to happen. It denotes the possibility of the outcome of any event.

Key Probability Terminology

- 1 Experiment: A repeatable procedure or action with an uncertain result (like rolling a die or tossing a coin).
- 2 Outcome: A single possible result of an experiment.
- 3 Sample Space(S): The complete set of all possible outcomes for an experiment.
- 4 Event(E): A specific subset or combination of outcomes for an experiment.
- 5 Favourable Outcome: An outcome that satisfies the specific condition of the event.
- 6 Complementary Event: The opposite of an event happening, the event not occurring where $P(A) + P(\bar{A}) = 1$.
- 7 Mutually Exclusive: Two or more events that cannot happen at same time.

8. Independent Events: Events where the outcome of one does not affect the probability of the other

Give atleast three probability event examples from the dataset.

Event A: a randomly chosen student passed the exam

$$P(A) = \frac{n(A)}{n(S)} = \frac{129}{200} = 0.6450$$

Event B: a randomly chosen student studies more than 10 hours per week

$$P(B) = \frac{n(B)}{n(S)} = \frac{95}{200} = 0.4750$$

Event C: a randomly chosen student has attendance above 80% & passed

$$P(C) = \frac{n(C)}{n(S)} = \frac{67}{200} = 0.3350$$

2 Types of Events.

Identify one empirical probability & one theoretical probability scenario from the dataset & calculate both

Empirical (experimental) probability

$$P(E) = \frac{\text{no. of times E occurred}}{\text{total no. of observations.}}$$

Output: Student passes the exam

$$P(\text{pass}) = 129/200 = 0.6450$$

Theoretical probability:

$$P(E) = \frac{\text{favourable outcomes}}{\text{total possible outcomes}}$$

Output: an uniformed guess on a 4-option mcq
 $P(\text{correct}) = 1/4 = 0.2500$

3. Random Variable & Probability Distribution.

Define a random variable for the event "No. of students passing the final exam" out of 3 randomly selected students.

Each student is a trial with two outcomes & a pass probability p estimated from the data. With 200 students & only 3 selected, treating the trials as independent is a good approximation, so X follows a Binomial ($n=3, p$) distribution.

$$P(X=x) = C(3, x) \cdot p^x \cdot (1-p)^{3-x}$$

X can take the values 0, 1, 2, 3.

$$p = P(\text{pass}) = 129/200 = 0.645$$

$$q = P(\text{fail}) = 1 - p = 0.355$$

(b) Construct its probability distribution table

x		
0	$C(3,0) \times 0.645^0 \times 0.355^3 =$	0.0447
1	$C(3,1) \times 0.645^1 \times 0.355^2 =$	0.2439
2	$C(3,2) \times 0.645^2 \times 0.355^1 =$	0.4431
3	$C(3,3) \times 0.645^3 \times 0.355^0 =$	0.2683
		1.0000

Find the mean & variance of this random variable

$$\begin{aligned}\text{Mean} &= E(X) = \sum x p(x) \\ &= (0 \times 0.447) + (1 \times 0.2439) + (2 \times 0.4431) + (3 \times 0.2683) \\ &= 0 + 0.2439 + 0.8862 + 0.8047 \\ &= 1.935\end{aligned}$$

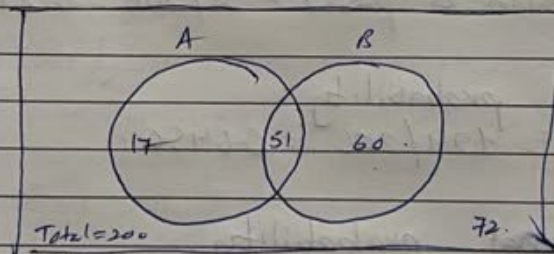
$$\begin{aligned}\text{Variance} &= \text{Var}(X) = E(X^2) - [E(X)]^2 \\ &= 0^2 \times 0.447 + (1^2 \times 0.2439) + (2^2 \times 0.4431) + (3^2 \times 0.2683) \\ &\quad - (1.935)^2 \\ &= 0.687\end{aligned}$$

4 Venn Diagram in Probability

A: Student who study more than 10 hours/week

B: Student who attend more than 80% of classes

Overlap of both



5 Contingency Table & Probability Calculations.

Create a contingency table for group discussion (Yes/No) vs final exam - pass

Calculate (i) Joint probability

(ii) Marginal

(iii) Conditional

Contingency table (counts)

final-exam-pass. group-discussion	Pass	Fail	Total
Yes	58	26	84
No	71	35	116
Total	129	71	200

Contingency table (joint probabilities = count/200)

final-exam-pass. group-discussion	Pass	Fail	Total
Yes	0.290	0.130	0.42
No	0.355	0.235	0.58
Total	0.645	0.355	1.00

(a) Joint probability

$$P(\text{discussion \& pass}) = 58/200 = 0.2900$$

(b) Marginal probability

$$P(\text{pass}) = 129/200 = 0.6450$$

(c) Conditional probability

$$P(\text{Pass/Discussion}) = \frac{58/200}{84/200} = \frac{58}{84} = 0.6905$$

6

Understanding Relationships

Interpret conditional probability results in plain language

Identify if participating in group discussion & passing exam are independent, dependent or mutually exclusive

$P(A/B) = P(A \text{ and } B) / P(B)$ mean: restrict attention to only the students for whom B is true & ask what fraction for them also have A. The denominator shrinks from all 200 students to just the B group.

- (i) mutually exclusive $P(A \text{ and } B) = 0$.
- (ii) independent $P(A \text{ and } B) = P(A) \times P(B)$ or $P(A/B) = P(A)$
- (iii) chi-square test of independence

$$P(\text{discussion}) = 0.4200$$

$$P(\text{Pass}) = 0.6450$$

- (i) $P(D \& P) = 0.2900$, which is not 0, so events are not mutually exclusive
- (ii) $P(D) \text{ and } P(P) = 0.4200 \times 0.6450 = 0.2709$
 $P(D \text{ and } P) = 0.2900$
 $P(D/P) = 0.6905$ versus $P(P) = 0.6450$
- (iii) chi square test of independence.
 $\chi^2 = 1.308$, $\text{dof} = 1$, $p\text{-value} = 0.2527$

7 Baye's Theorem Application

- 70% of students who pass has high attendance
- 40% of students who fail also had high attendance
- 60% of students had high attendance

$$P(\text{High attendance} / \text{Pass}) = 0.70.$$

$$P(\text{High attendance} / \text{Fail}) = 0.40$$

$$P(\text{High attendance}) = 0.60$$

$$P(\text{Pass} / \text{High attendance}) = ?$$

$$P(\text{Fail}) = 1 - P(\text{Pass})$$

$$P(H) = P(H / \text{Pass}) \times P(\text{Pass}) + P(H / \text{Fail}) \times (1 - P(\text{Pass}))$$

$$0.6 = 0.7 \times p + 0.4 (1 - p)$$

$$p = \frac{(0.6 - 0.4)}{0.7 - 0.4} = 0.6667 = \frac{2}{3}$$

$$P(\text{Pass}) = 0.6667$$

$$P(\text{Fail}) = 0.3333$$

$$\begin{aligned} P(\text{Pass} / H) &= P(H / \text{Pass}) \times P(\text{Pass}) / P(H) \\ &= 0.7 \times 0.6667 / 0.6 \\ &= 0.7778 \end{aligned}$$

$$P(\text{Fail} / H) = 0.4 \times 0.3333 / 0.6 = 0.2222$$

$$P(\text{Pass} / H) = 0.7778 \quad P(\text{Fail} / H) = 0.2222$$